Coherent response and residual fluctuations in a light-modulated circadian oscillator

Liyan Gong, Charles S. Peskin

CIMS, NYU

AML Meeting October 16, 2026

Background: models of the circadian clock

Forger and Peskin: detailed mammalian clock models

- **2003:** A detailed biochemical model to reproduce circadian mRNA and protein rhythms.
- **2005:** A stochastic formulation to study clock precision and robustness to molecular noise.

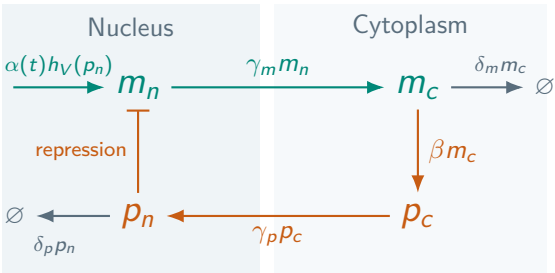
Wang and Peskin (2018): a simpler cellular oscillator

- One gene and four molecular species: nuclear and cytoplasmic mRNA and protein.
- Negative feedback generates oscillations. Periodic light modulates transcription.
- Independently control the intrinsic period and relative noise; vary light modulation at a fixed mean light level.

Our work extends Wang and Peskin (2018) to quantify the protein response.

Forger and Peskin (2003), *PNAS* **100**, 14806–14811. doi:10.1073/pnas.2036281100.
Forger and Peskin (2005), *PNAS* **102**, 321–324. doi:10.1073/pnas.0408465102.
Wang and Peskin (2018), *Physical Review E* **97**, 062416. doi:10.1103/PhysRevE.97.062416.

A negative-feedback circuit generates the oscillation



Model adapted from Wang and Peskin (2018).

mRNA: nuclear and cytoplasmic: m_n and m_c.

Protein: cytoplasmic and nuclear: p_c and p_n.

Six reaction channels define the stochastic model

Molecule counts: $\mathbf{N} = (m_n, m_c, p_c, p_n)^{\mathsf{T}}$.

j	Reaction	Rate $a_j(\mathbf{N}, t)$	State change
1	Transcription	$\alpha(t) h_V(p_n)$	$m_n \to m_n + 1$
2	mRNA export	$\gamma_m m_n$	$m_n \to m_n - 1, \quad m_c \to m_c + 1$
3	mRNA degradation	$\delta_m m_c$	$m_c \to m_c - 1$
4	Translation	βm_c	$p_c \to p_c + 1$
5	Protein import	$\gamma_p p_c$	$p_c \to p_c - 1, \quad p_n \to p_n + 1$
6	Protein degradation	$\delta_p p_n$	$p_n \to p_n - 1$

$$h_V(p_n) = P_0 = \left[\sum_{\ell=0}^{\min(p_n, r)} \binom{r}{\ell} \frac{p_n!}{(p_n - \ell)!\,(KV_n)^{\ell}} \right]^{-1}.$$

P_0 is the probability that all r binding sites are unoccupied. Here K is the dissociation constant in concentration units, and V_n is the nuclear volume.

Wang and Peskin (2018), Table I and Eq. (5). The translation channel creates protein without consuming mRNA.

Concentrations and deterministic dynamics

$$M_n = \frac{m_n}{V_n}, \qquad M_c = \frac{m_c}{V_c}, \qquad P_c = \frac{p_c}{V_c}, \qquad P_n = \frac{p_n}{V_n}.$$

V_n and V_c are the nuclear and cytoplasmic volumes.

Concentration equations

$$\begin{aligned}
\dot{M}_n &= \frac{\alpha(t)}{V_n} h(P_n) - \gamma_m M_n, \\
\dot{M}_c &= \gamma_m \frac{V_n}{V_c} M_n - \delta_m M_c, \\
\dot{P}_c &= \beta M_c - \gamma_p P_c, \\
\dot{P}_n &= \gamma_p \frac{V_c}{V_n} P_c - \delta_p P_n.
\end{aligned}$$

Wang and Peskin (2018), Eqs. (9)–(13).

Macroscopic limit

$$h(P_n) = \left(\frac{K}{K + P_n} \right)^r.$$

For $p_n \gg r$,

$$h_V(p_n) \simeq h(p_n / V_n).$$

Periodic light input

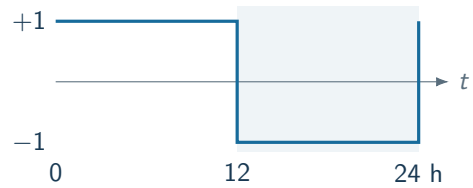
$$\boxed{\alpha(t) = \alpha_0 + \bar{\alpha}[1 + \epsilon_0 s(t)]}.$$

Transcription parameters

- α_0: maximal transcription rate in complete darkness.
- $\bar{\alpha}$: mean light-induced increment to this rate.
- ϵ_0: depth of light modulation.

Our simulations: $\bar{\alpha} = 1.5\alpha_0$.

The square wave $s(t)$



Light period: $\tau_L = 24$ h.

Dark free-running period: $T_0 = 23.2$ h.

Varying ϵ_0 changes the modulation depth at fixed mean light input.

Light-input form: Wang and Peskin (2018), Eq. (36).

Parameters

Volume scaling and light amplitude

$$s_V \in \{1, 2, 4, 8\},$$
$$\epsilon_0 \in \{0.2, 0.4, 0.6, 0.8, 1.0\}.$$

Volume scaling means $(V_n, V_c) = s_V(V_{n,0}, V_{c,0})$.
Sixteen independent trajectories per case.

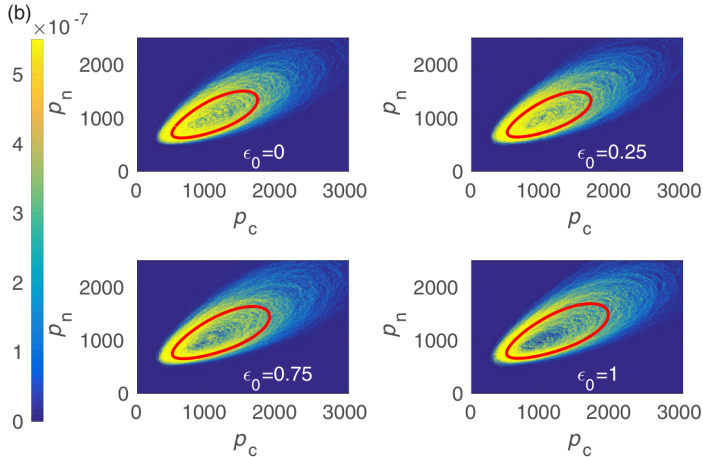
600 days simulated, first 100 discarded.

Sampling interval: 0.5 h, giving 48 light phases.

Observable

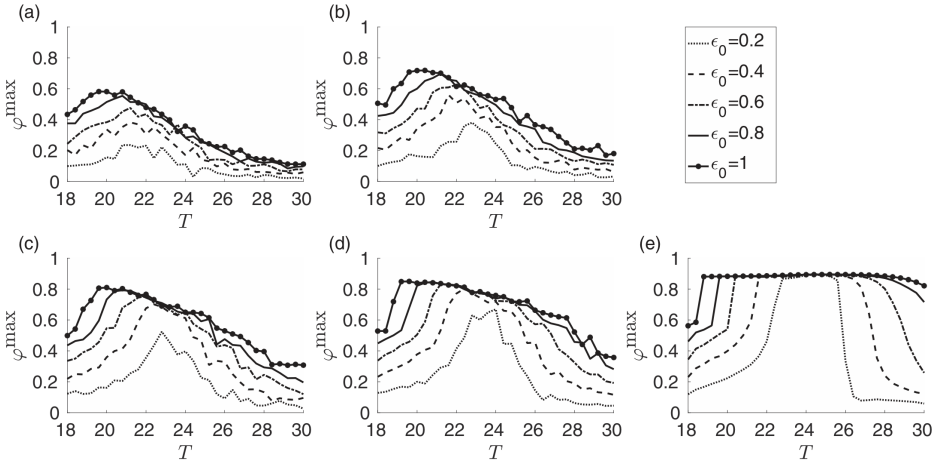
$$F(t) = p_c(t) + p_n(t).$$

Motivation: organization of protein states



Wang and Peskin (2018), Fig. 7(b).

Motivation: protein–light correlation



Wang and Peskin (2018), Fig. 8.

Motivation: questions for our extension

> *"We have not quantified this phenomenon,*
> *and we certainly have not explained it."*

Wang and Peskin (2018), Conclusions, p. 10: the organizing effect in Fig. 7.

Our extension asks three related questions:

- Why does stronger light modulation increase protein–light correlation?
- How much of the residual protein variability reflects differences along the oscillation cycle, and how much reflects deviations away from it?
- Which reactions contribute to protein variability, and how do local fluctuations propagate from one light cycle to the next?

Coherent and residual protein variability

Pool observations uniformly over the 24-hour light cycle.

$$q = t \bmod \tau_L, \qquad \mu_F(q) = \mathbb{E}[F \mid q].$$

Coherent variance

$$S = \mathrm{Var}[\mu_F(q)].$$

Variation of the mean response across light phases.

Residual variance

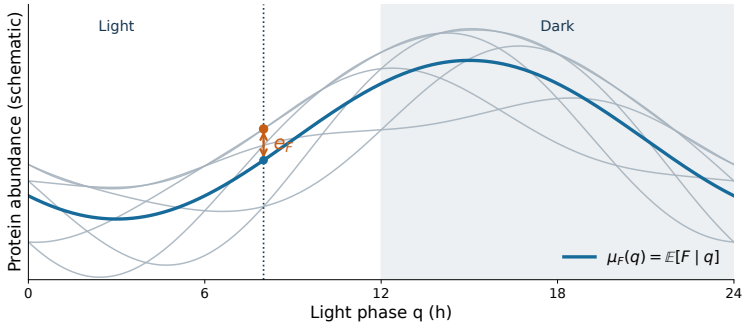
$$N = \mathbb{E}[\mathrm{Var}(F \mid q)].$$

Variability at a fixed light phase, averaged over phase.

$$\mathrm{Var}(F) = S + N, \qquad \boxed{\mathcal{C}_{24} = \frac{S}{S+N}}.$$

$\mathcal{C}_{24}$ is the fraction of total protein variance in the repeatable response.

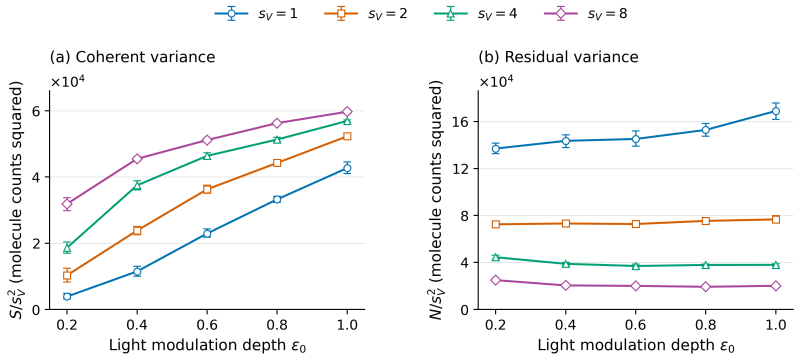
Conditioning on light phase separates the repeatable response



$$q = t \bmod \tau_L, \qquad F = \underbrace{\mu_F(q)}_{\mathbb{E}[F|q]} + e_F.$$

Schematic illustration of the definition.

Stronger forcing primarily increases the coherent response S



S increases at every volume. The response of N depends on volume.

Variances are divided by s_V^2 for comparison.

Two factors determine protein–light correlation

Let $L_\ell(q) = L(q + \ell)$ denote light shifted by lag ℓ, $\quad 0 \le \ell < 24\text{h}$.

$$\Phi_{\text{period}} = \max_{\ell} \text{Corr}\big(F, L_\ell(q)\big),$$

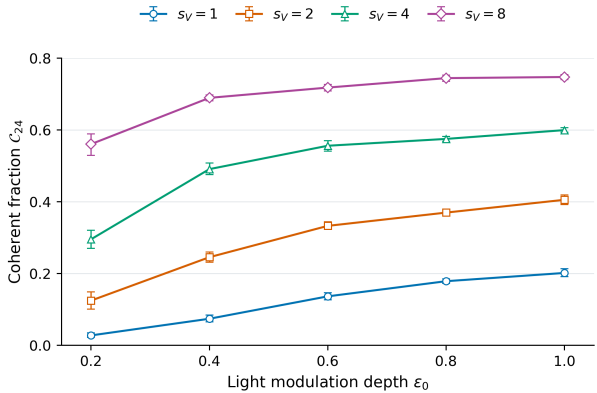
$$R_{\text{shape}} = \max_{\ell} \text{Corr}_q\big(\mu_F(q), L_\ell(q)\big).$$

- R_{shape}: matching of the mean waveform to light.
- $\mathcal{C}_{24}$: fraction of variance in the repeatable response.

$$\boxed{\Phi_{\text{period}} = R_{\text{shape}}\sqrt{\mathcal{C}_{24}}}.$$

Correlation depends on both waveform matching and the coherent fraction.

The coherent fraction $\mathcal{C}_{24}$ explains the correlation Φ_{period} increase



Answer: Why protein–light correlation increases

$$\Phi_{\text{period}} = R_{\text{shape}}\sqrt{\mathcal{C}_{24}} = R_{\text{shape}}\sqrt{\frac{S}{S+N}}.$$

Across the twenty cases,

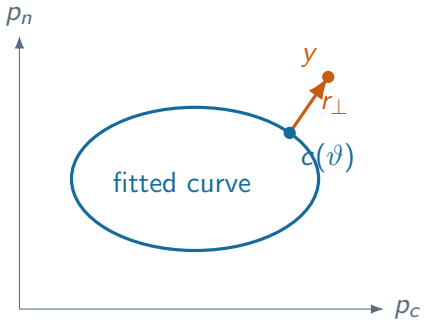
$$R_{\text{shape}} \in [0.8964, 0.8986].$$

As light modulation increases: the increase of $\mathcal{C}_{24}$ implies an increase in S or decrease in N:

- **The repeatable response S becomes stronger.**
  S increases: the average protein response varies more strongly over the light cycle.
- **Residual variability N does not consistently decrease.**
  Changes in N depend on cell volume and are smaller relative to the growth in S.

Higher correlation mainly reflects a stronger repeatable response throughout the day.

Residual variation



Schematic projection

$$y = (p_c, p_n)^\mathsf{T} = c(\vartheta) + r_\perp.$$

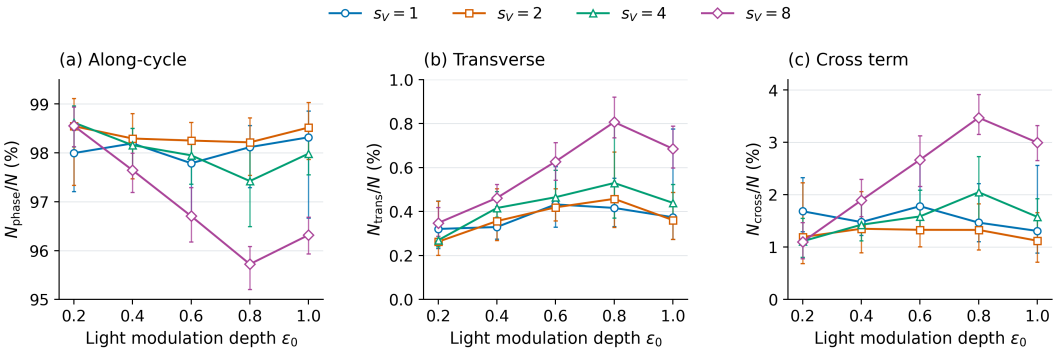
$$F = \underbrace{(1,1)c(\vartheta)}_{g_F(\vartheta)=c_n(\vartheta)+c_c(\vartheta)} + \underbrace{(1,1)r_\perp}_{h_F}.$$

$$N_{\text{phase}} = \mathbb{E}_q \operatorname{Var}(g_F \mid q),$$
$$N_{\text{trans}} = \mathbb{E}_q \operatorname{Var}(h_F \mid q),$$
$$N_{\text{cross}} = 2\mathbb{E}_q \operatorname{Cov}(g_F, h_F \mid q).$$

$$\boxed{N = N_{\text{phase}} + N_{\text{trans}} + N_{\text{cross}}}.$$

Motivated by the protein-state cycle in Wang and Peskin (2018), Fig. 7(b), we fit a closed reference curve to our Gillespie protein states. Each state y is projected onto its nearest point $c(\vartheta)$ on this curve.

Along-curve variation accounts for most of the residual



$N_{\text{phase}}/N = 95.7$–$98.6\%$ **across the primary fits.**

Which numerical approximations preserve the observables

Method	Reaction update	Role in the comparison
Gillespie	One random reaction at a time	Reference jump process
Chemical Langevin	Gaussian increments	Diffusion approximation
Hybrid	mRNA jumps, protein ODE	Deterministic protein reactions
Poisson tau-leap	Poisson counts in each step	Random counts in all channels

Assess mean concentrations, covariance, and protein–light correlation.

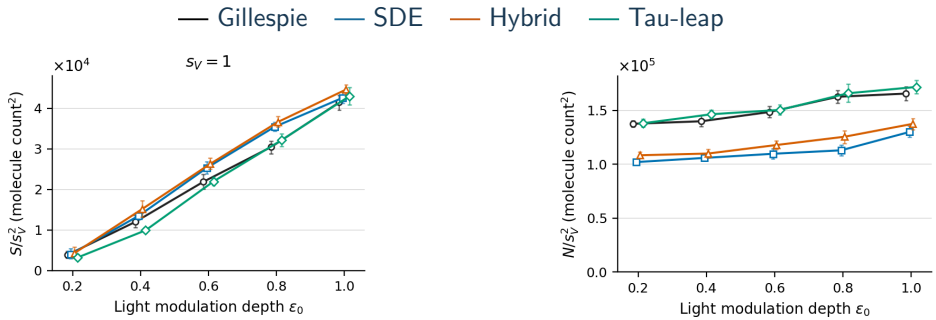
Tau-leaping has the smallest average errors

Errors relative to Gillespie, averaged across the twenty cases.

Statistic	SDE	Hybrid	Tau-leap
Mean-vector error E_μ (%)	1.507	1.072	0.321
Covariance error E_Σ (%)	9.476	16.707	3.341
Correlation error E_ϕ	0.02055	0.03605	0.00699

E_μ and E_Σ are relative norm errors; E_ϕ is an absolute correlation difference.

Comparing coherent and residual variance



The methods agree more closely on S than on N. Poisson tau-leaping stays close to Gillespie for both in this comparison.

The linear noise approximation (LNA)

Let $x_*(t)$ be a stable τ_L-periodic solution of $\dot{x} = f(x,t)$, and write $\xi \simeq x - x_*$.
Linearizing the deterministic drift around this orbit gives

$$f(x_*(t)+\xi,t) - f(x_*(t),t) \simeq J(t)\xi, \qquad J(t) = D_x f(x_*(t),t).$$

$$d\xi = J(t)\xi\, dt + \sum_{j=1}^{6} B_j(t)\, dW_j(t).$$

Propagation

$J(t)$ describes the local deterministic response. Apply Itô's product rule to obtain the covariance equation:

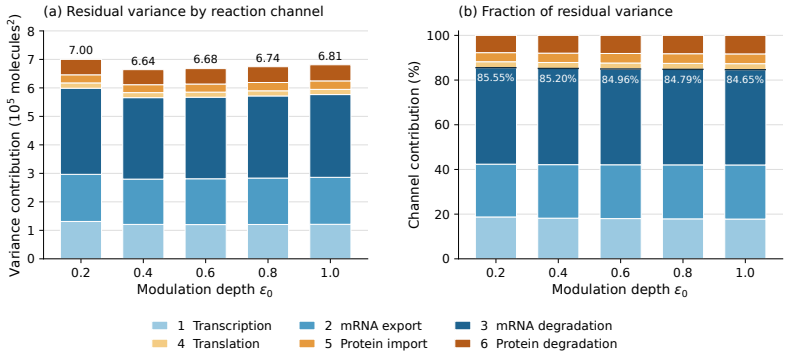
Noise injection

$D_j(t) = B_j(t)B_j(t)^{\mathsf{T}}$ describes reaction channel j, where $B_j(t)$ is the noise amplitude.

$$\boxed{\dot{C} = JC + CJ^{\mathsf{T}} + \sum_j D_j}, \qquad C(t+\tau_L) = C(t).$$

Independent Wiener processes W_j. The local calculation uses the macroscopic reaction rates along the deterministic orbit.

mRNA channels dominate the local predicted protein variance



$$C = \sum_j C_j, \qquad N_{\mathrm{LNA},j} = \mathbb{E}[b^{\mathsf{T}} C_j(q) b], \qquad b = (0, 0, V_c, V_n)^{\mathsf{T}}.$$

Reaction-channel decomposition: Komorowski, Miękisz and Stumpf (2013). mRNA channels contribute 84.65–85.55% within the periodic LNA.

A daily linear map

The homogeneous perturbation equation is $\dot{\delta x} = J(t)\delta x$. By linearity, its solution has the form $\delta x(t) = \Phi(t,s)\delta x(s)$, where $\Phi(t,s)$ propagates perturbations from time s to t. Differentiating gives $\partial_t \Phi(t,s) = J(t)\Phi(t,s)$, with $\Phi(s,s) = I$.

$$\xi(t) = \Phi(t,s)\xi(s) + \sum_j \int_s^t \Phi(t,u) B_j(u)\, dW_j(u).$$

$$\xi_n = \xi(t_0 + n\tau_L), \qquad \Psi = \Phi(t_0 + \tau_L, t_0).$$

$$\boxed{\xi_{n+1} = \Psi \xi_n + \zeta_n}, \qquad \zeta_n \sim \mathcal{N}(0, Q).$$

$$Q = \int_{t_0}^{t_0+\tau_L} \Phi(t_0 + \tau_L, s) D(s) \Phi(t_0 + \tau_L, s)^{\mathsf{T}}\, ds.$$

A left eigenvector produces a closed linear readout

We seek a complex coordinate $u_n = \ell^{\mathsf{T}} \xi_n$.

$$u_{n+1} = \ell^{\mathsf{T}} \Psi \xi_n + \ell^{\mathsf{T}} \zeta_n.$$

Choose ℓ so that $\ell^{\mathsf{T}} \Psi = \lambda \ell^{\mathsf{T}}$, with $\lambda = c + id$, $d > 0$.

$$u_{n+1} = \lambda \underbrace{\ell^{\mathsf{T}} \xi_n}_{u_n} + \ell^{\mathsf{T}} \zeta_n.$$

$$\boxed{u_{n+1} = \lambda u_n + \ell^{\mathsf{T}} \zeta_n}.$$

T denotes transpose, including for the complex vector.

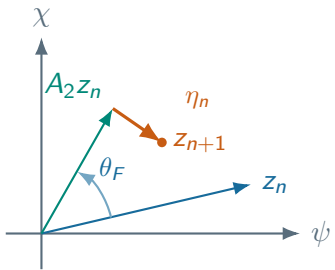
The two real coordinates rotate and contract each day

Write $u_n = \psi_n + i\chi_n$ and $z_n = (\psi_n, \chi_n)^{\mathsf{T}}$.

$$z_{n+1} = \underbrace{\begin{pmatrix} c & -d \\ d & c \end{pmatrix}}_{A_2 = \rho_F R(\theta_F)} z_n + \eta_n.$$

$$\rho_F = |\lambda|, \qquad \theta_F = \arg\lambda.$$

For $\rho_F < 1$, old perturbations decay.



Schematic map

Existing deviations rotate and shrink, while new noise enters each cycle.

Floquet analysis of noisy oscillators: Boland, Galla and McKane (2009).

A local timing coordinate in radians

The complex eigenvector has arbitrary scale and phase. We calibrate its readout so that a small timing shift of $\delta\psi$ radians gives $(\psi, \chi) \simeq (\delta\psi, 0)$.

$$g = \frac{\dot{x}_*(t_0)}{\omega_L}, \qquad \omega_L = \frac{2\pi}{\tau_L}.$$

A tangent displacement of $\delta\psi$ radians has $\xi \simeq g\,\delta\psi$.

$$\ell^{\mathsf{T}} g = 1, \qquad W_2 = [\Re\ell, \Im\ell], \qquad z = W_2^{\mathsf{T}} \xi.$$

$$W_2^{\mathsf{T}} g = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad z \simeq \begin{pmatrix} \delta\psi \\ 0 \end{pmatrix}.$$

ψ reads a small timing displacement; χ is its coupled Floquet coordinate.

Assumes $\ell^{\mathsf{T}} g \neq 0$ before rescaling.

Noise input and stationary covariance

The projected daily map is

$$z_{n+1} = A_2 z_n + \eta_n, \qquad z_n = W_2^{\mathsf{T}} \xi_n, \qquad \eta_n = W_2^{\mathsf{T}} \zeta_n.$$

Noise accumulated during one cycle

$$Q_2 := \mathrm{Cov}(\eta_n) = W_2^{\mathsf{T}} Q W_2.$$

Here $Q_2 = \mathrm{Cov}(\zeta_n)$ is the covariance of the new noise accumulated during one cycle and projected into the two coordinates.

Covariance of the current state

$$P_2 := \mathrm{Cov}(z_n).$$

This covariance P_2 is the covariance of current state, and it includes fluctuations retained from previous cycles.

Since η_n is independent of z_n,

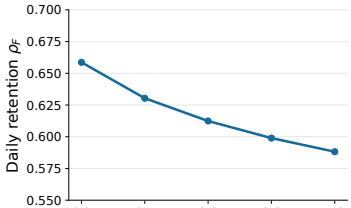
$$P_{2,n+1} = A_2 P_{2,n} A_2^{\mathsf{T}} + Q_2.$$

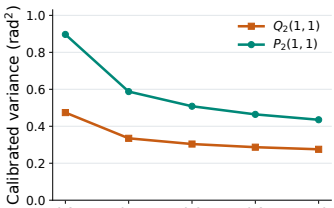
For $\rho(A_2) < 1$, the covariance approaches a stationary value:

$$\boxed{P_2 = A_2 P_2 A_2^{\mathsf{T}} + Q_2}.$$

All covariances refer to the fixed sampling phase t_0. At stationarity, $P = \mathrm{Cov}(\xi_n) = C(t_0)$ and $P_2 = W_2^{\mathsf{T}} P W_2$.

Local contraction and predicted timing variance





For the map $z_{n+1} = A_2 z_n + \eta_n$, with $z_n = (\psi_n, \chi_n)^\mathsf{T}$:

$$Q_2(1,1) = \mathrm{Var}(\eta_{\psi,n})$$

New noise generated during one cycle and propagated to its end.

$$P_2(1,1) = \mathrm{Var}(\psi_n)$$

Stationary local timing variance, including noise retained from earlier cycles.

$s_V = 4$. Here $\eta_{\psi,n}$ is the first component of η_n. Both variances are LNA predictions at the fixed sampling phase t_0, expressed in radians squared.

Stronger light modulation gives faster recovery and smaller day-to-day timing fluctuations.

$$\underbrace{P_2}_{\text{stationary covariance}} = \underbrace{A_2 P_2 A_2^{\mathsf{T}}}_{\text{fluctuations retained from yesterday}} + \underbrace{Q_2}_{\text{new daily noise}} .$$

- **Stronger local contraction:** $\rho_F \downarrow$.
  Without new noise, perturbations in the Floquet coordinates decay faster from one day to the next.

- **Smaller daily noise contribution:** $Q_2(1,1) \downarrow$.
  Noise accumulated and propagated over one cycle produces less variability in the calibrated timing coordinate.

- **Smaller stationary timing variance:** $P_2(1,1) \downarrow$.
  The timing coordinate fluctuates less when observed at the same light phase each day.

Take-home messages

1. **Correlation:** A stronger coherent response raises $\mathcal{C}_{24}$ and correlation, while waveform matching changes little.
2. **Organization:** Along-curve variation accounts for 95.7–98.6% of the residual protein variance.
3. **Noise dynamics:** Within periodic LNA, mRNA channels contribute about 85% of residual protein variance. Stronger forcing reduces daily retention and local timing variance.

Discussion

Next question

How do these components change near entrainment boundaries, where phase slips become important?

Acknowledgments

Charles S. Peskin: research guidance and collaboration.

Yuri Bakhtin: the hybrid-method suggestion and discussions of the daily LNA map.

Leif Ristroph and AML group members: encouragement and valuable suggestions for my talk.

Thank you! Any questions?